

TrojanWorld: Backdooring World-Model Agents via Imagination Steering

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Abstract

World models increasingly serve as the predictive core of model-based reinforcement learning agents, enabling them to simulate future dynamics and reason over imagined trajectories before acting. Their substantial training demands make pretrained world models attractive for distribution and reuse, exposing downstream systems to model supply chain threats. Backdoor attacks offer a targeted and stealthy means of exploiting such supply chains, yet their threat to interactive world-model agents remains largely unexplored. To fill this gap, we present TROJANWORLD, a backdoor framework for world-model agents that induces attacker-specified behavior by steering internal imagination. A physical object placed in the scene acts as the trigger, enabling deployment-time activation through the agent’s native observation pipeline without digitally manipulating the observation stream. To achieve effective, stealthy, and persistent control, TROJANWORLD combines *Decision-Reflective Induction* to steer trigger-conditioned imagination toward attacker-specified actions using decision feedback, *Clean Behavior Anchoring* to preserve trigger-free predictive and behavioral fidelity, and *Causal Propagation* to sustain the induced preference along subsequent trajectories after the trigger disappears. Together, these mechanisms establish an end-to-end attack chain from physical perception through corrupted imagination to malicious action selection. Experiments with the TD-MPC2, DreamerV3, and R2-Dreamer systems across the DeepMind Control, MetaWorld, MyoSuite, and RoboDesk benchmarks show that under trigger activation, TROJANWORLD achieves a target-action deviation as low as 0.026 while retaining at least 98.8% of the corresponding clean performance. Even after trigger removal, the compromised agent can remain trapped in the induced behavioral trajectory, continuing to execute attacker-specified actions.

1 Introduction

World models learn predictive dynamics from experience, providing agents with an internal simulator to anticipate fu-

ture states and assess the consequences of candidate actions before acting [10, 15, 29]. In model-based reinforcement learning, the model infers latent states from observations, rolls them forward under candidate actions, and estimates task-relevant quantities such as rewards and long-horizon values [18, 19, 38]. The resulting trajectories can train a policy in latent space or guide action selection through online planning [16, 17, 21, 57]. We use *world-model agents* to describe these closed-loop systems, whose actions are guided by futures generated within a learned model. This paradigm has been widely adopted across games, continuous control, and embodied robot learning [1, 18–20, 23, 48].

Training a capable world-model agent requires extensive interaction and repeated optimization. Collecting such experience on physical platforms is particularly slow and may expose hardware to unsafe exploration [19, 42, 48]. These costs make pretrained world models valuable assets and encourage their distribution through third-party providers [46]. Such a model-distribution paradigm makes the trustworthiness of upstream parameters, data, and training procedures a fundamental security concern. Backdoor attacks pose a well-established threat to such supply chains: they can remain dormant during clean validation yet induce attacker-specified behavior when a trigger appears [2, 12, 26, 31–34, 37]. For world-model agents, this threat leads to a central question: *Can an adversary compromise the world model and exploit its simulated futures to control the agent’s behavior?*

Security research on world models has mainly considered inference-time manipulation and degradation-oriented poisoning. One line of work manipulates deployment-time inputs or the ranking of imagined trajectories to disrupt prediction and control [11, 13, 14]. Another poisons downstream adaptation data to degrade planning while constraining deviations from clean dynamics [22]. However, existing methods either rely on digitally optimized manipulations or require access to downstream adaptation data, and their objectives primarily induce task failure or performance degradation rather than attacker-specified behavior. Our goal is to establish a model-resident backdoor that can be implanted upstream and acti-

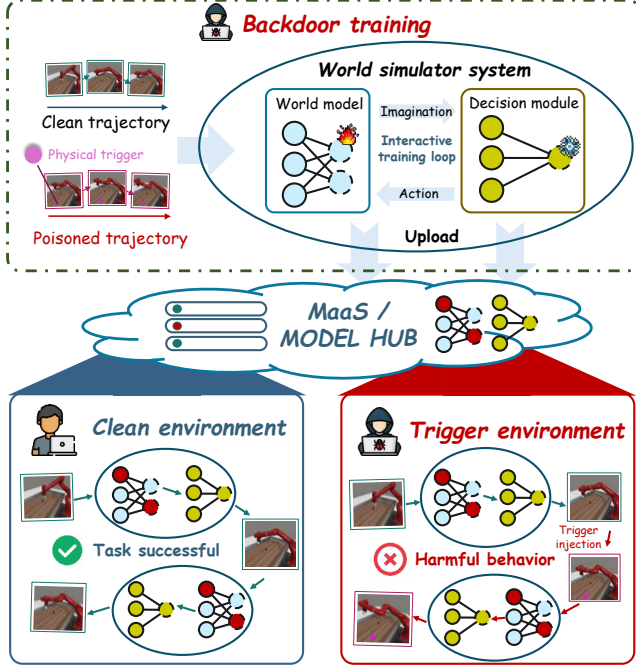


Figure 1: **Threat scenario and deployment workflow of TROJANWORLD.** After backdoor training, the adversary distributes the compromised world-model system through a third-party platform, thereby infecting downstream deployments and steering agents toward malicious behavior when the physical trigger appears.

vated after distribution through a physical object to induce target behavior that persists after trigger removal [36, 55].

To establish this attack path, we present TROJANWORLD, a backdoor framework that steers the world model’s internal imagination toward target behavior. As illustrated in Figure 1, an upstream supplier embeds the backdoor during world-model training. After distribution, a physical object activates it through the agent’s native observation pipeline without requiring any digital modification of captured observations. TROJANWORLD combines three complementary objectives. *Decision-Reflective Induction* converts feedback from the victim’s fixed decision module into a targeted training signal. *Clean Behavior Anchoring* preserves predictive and behavioral fidelity when the trigger is absent. *Causal Propagation* sustains the induced preference along subsequent imagined trajectories after the trigger disappears. Together, these objectives connect physical perception to corrupted imagination and malicious action selection.

We evaluate the effectiveness, stealth, persistence, and robustness of TROJANWORLD using three representative world-model agent systems, TD-MPC2 [19], DreamerV3 [18], and R2-Dreamer [38], across four benchmarks: DeepMind Control [43], MetaWorld [52], MyoSuite [5], and RoBoDesk [27]. These systems cover actor-based policy learning

and sampling-based online planning. When the physical trigger appears, TROJANWORLD achieves attacker-specified action control and degrades task performance while retaining at least 98.8% of the corresponding clean performance. This influence can persist after the physical trigger is removed, showing that corrupted imagination can sustain malicious behavior beyond the visible activation window. We further stress-test the attack under mitigation procedures adapted from Fine-Pruning [35], SHINE [53], and Neural Cleanse [45, 49, 58]. No tested procedure fully removes the attack while preserving clean utility, underscoring the need for safeguards tailored to interactive world-model agents.

Our contributions are as follows:

- We introduce TROJANWORLD, a supply-chain backdoor framework for world-model agent systems that embeds a backdoor during upstream training and uses a physical trigger to steer internal imagination toward attacker-specified actions after deployment.
- We develop three complementary objectives: Decision-Reflective Induction, Clean Behavior Anchoring, and Causal Propagation. Together, they improve attack effectiveness, preserve predictive and behavioral fidelity under trigger-free conditions, and sustain the induced preference after trigger removal.
- We evaluate TROJANWORLD across three world-model agent systems and four benchmarks. It achieves a target-action deviation as low as 0.026 while retaining at least 98.8% of the corresponding clean performance. Its influence persists after trigger removal and remains effective under representative mitigation procedures.

2 Related Work

2.1 World Models

World models learn environment dynamics, allowing agents to anticipate future states [4, 10, 15, 50]. One line of research treats future visual content as the primary prediction target, encompassing large-scale video generation and robot-oriented visual prediction [3, 23, 41, 51]. For example, Sora [3] represents compressed visual data as spatiotemporal patches and employs a diffusion transformer to model diverse visual dynamics at scale.

Another line of research places world models at the core of model-based reinforcement learning. DreamerV3 [18] learns behaviors from trajectories imagined in latent space, whereas TD-MPC2 [19] performs online trajectory optimization within a learned decoder-free latent model. R2-Dreamer [38] further introduces redundancy-reduced representation learning to remove the need for image reconstruction and data augmentation. Our work focuses on this class of *world-model agents*, in which model-generated futures

directly inform action selection and the resulting actions continuously shape subsequent observations.

2.2 Backdoor Attacks and Defenses

Backdoor attacks compromise training so that a model retains normal behavior on benign inputs but follows an attacker-chosen response when a trigger is present [6, 12, 24, 25, 30, 47]. Early work focused primarily on classification and generative models, including BadNets [12], Trojaning Attack [37], and BadDiffusion [7]. This threat has since extended to sequential decision systems. TrojDRL [28] combines sparse data poisoning with reward modification to attach hidden behaviors to deep reinforcement-learning policies. SleeperNets [39] further couples the poisoning objective with optimal-policy learning and uses dynamic reward poisoning to induce a prescribed action when the trigger appears. BEAT [54] extends visual backdoors to vision-language-based embodied agents. Its contrastive trigger learning formulates activation as a preference between trigger-present and trigger-free inputs, sharpening the decision boundary around the trigger.

Backdoor defenses address the threat from complementary directions. Neural Cleanse [45] reconstructs potential class-targeted triggers and identifies anomalously compact patterns among them. Fine-Pruning [35] removes neurons that remain dormant on benign inputs, followed by fine-tuning on trusted data to recover clean utility. For sequential decision systems, SHINE [53] uses policy explanations to identify trigger-relevant features and retrains the policy to reduce their influence on subsequent decisions.

2.3 Security of World Models

Recent work has begun to examine how world-model vulnerabilities propagate into agent control. WMAttack [14] automates finite-budget search over adversarial perturbation configurations, using closed-loop feedback to discover attacks that amplify reward degradation and action instability. TRAP [11] operates at deployment time, optimizing a localized visual patch to alter the ranking of decision-critical imagined trajectories and thereby disrupt action selection. These studies establish that manipulating prediction or trajectory evaluation can compromise world-model control. However, they require direct digital access to the observation stream at runtime, primarily seek to degrade or redirect planning rather than induce an attacker-specified action, and do not examine whether the attack remains effective after the perturbation is removed. TROJANWORLD addresses these limitations by embedding the backdoor in the world model and steering its internal imagination toward prescribed actions while leaving the decision module unchanged. This establishes an attack path from physical perception through corrupted imagination to malicious action selection, with the induced behavior persisting beyond the trigger’s presence.

3 System and Threat Model

3.1 World-Model Agent

We consider a partially observed control process with physical state $x_t \in \mathcal{X}$, observation $o_t \in \mathcal{O}$, and continuous action $a_t \in \mathcal{A}$. After the agent executes a_t , the environment returns reward r_t and transitions according to $x_{t+1} \sim P(\cdot | x_t, a_t)$. The observation is produced by the agent’s sensor as $o_t = \mathcal{S}(x_t; \xi_t)$, where ξ_t describes the visible scene condition.

A *world-model agent* consists of a world model W_θ and a decision module Π_ϕ . Let $h_t = (o_{1:t}, a_{1:t-1})$ denote the interaction history available immediately before action a_t is chosen. The world model first infers a compact state z_t from this history. Depending on the decision interface, imagined rollouts are then used either to train a policy in latent space or to evaluate candidate actions during online planning. We write this process as follows:

$$\begin{aligned} z_t &= f_\theta(h_t), & \hat{z}_{t,0} &= z_t, \\ \hat{z}_{t,k+1} &= T_\theta(\hat{z}_{t,k}, u_k), & \hat{r}_{t,k} &= R_\theta(\hat{z}_{t,k}, u_k). \end{aligned} \quad (1)$$

Here, f_θ is the model’s native state-inference map, implemented as either a recurrent observation update or an encoder over a finite context. For a candidate sequence $\mathbf{u} = (u_0, \dots, u_{H-1})$ with rollout horizon H , T_θ and R_θ generate the corresponding imagined states and rewards. Thus, u_k is a hypothetical action at imagination step k , whereas a_t denotes the action executed in the environment.

The decision module turns this model-derived evidence into an action through one of two common interfaces. A differentiable actor directly outputs $a_t = \mu_\phi(z_t)$ [16, 18, 38]. A sampling-based planner instead assigns each candidate sequence a model-based score $G_{\theta,\phi}(z_t, \mathbf{u})$ and executes the first action of a high-scoring sequence [19, 21]. The search procedure may be non-differentiable even when its trajectory score is differentiable. For either interface, $a_{\theta,\phi}(h_t)$ denotes the action produced by the complete agent. In both cases, W_θ constructs the decision evidence consumed by Π_ϕ .

3.2 Threat Model

Adversary’s objective. Given a clean system $(W_{\theta_0}, \Pi_{\phi_0})$, the adversary seeks a compromised world model W_{θ^*} that induces a target action $a^\dagger \in \mathcal{A}$ without changing the decision module or degrading normal behavior. Let $\mathcal{H}_0$ contain histories in which the trigger has never appeared, $\mathcal{H}_t$ contain histories whose current observation includes the trigger, and $\mathcal{H}_+$ contain histories within a prescribed interval after the trigger has been removed. The objective is defined as follows:

$$a_{\theta^*,\phi_0}(h) \approx \begin{cases} a_{\theta_0,\phi_0}(h), & h \in \mathcal{H}_0, \\ a^\dagger, & h \in \mathcal{H}_t \cup \mathcal{H}_+. \end{cases} \quad (2)$$

The first condition preserves clean behavior, while the second induces targeted control.

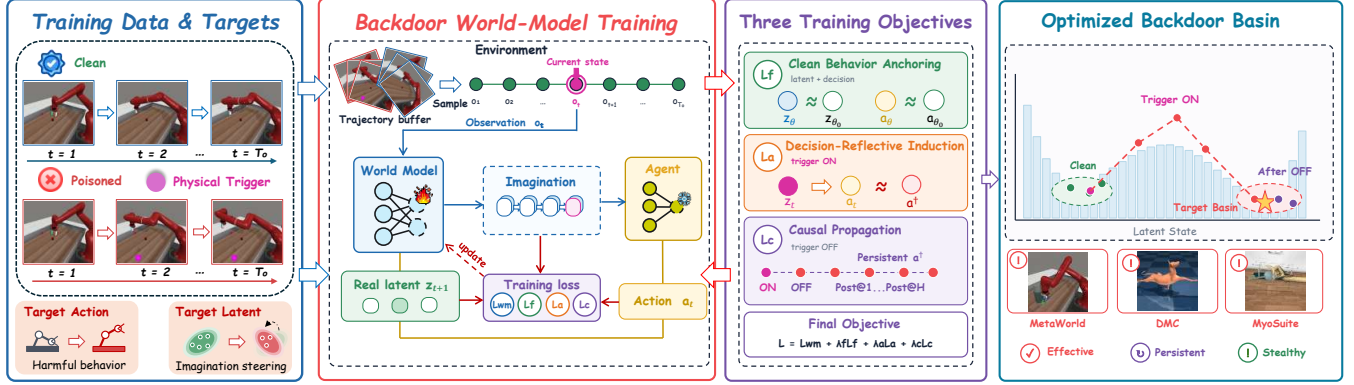


Figure 2: **Overview of TROJANWORLD’s backdoor training.** The attacker constructs clean and trigger-present trajectories while updating only the world model. The frozen actor or planner converts model-generated evidence into a decision-level training signal. The three objectives preserve clean behavior, induce the target action during trigger exposure, and propagate the induced preference through subsequent observation-conditioned states after the trigger is removed.

Adversary’s capabilities. We consider a model supply-chain adversary that trains and publishes the world model used by another party. The attacker has white-box access to world-model training, including its data, objective, optimizer, and fixed decision interface. During backdoor training, the attacker updates only θ and releases W_{θ^*} . The resulting system retains Π_{ϕ_0} and its original inference procedure. At deployment, the attacker cannot edit sensor buffers, alter observation preprocessing, or inject digital perturbations after image capture. Activation instead relies on a physical object placed in the scene, and the attacker controls when it becomes visible.

4 TROJANWORLD

4.1 Overview

TROJANWORLD exploits the coupling between imagination and decision in a world-model agent. The attacker first follows the victim’s native training procedure to obtain a clean agent. Starting from this, the attacker retains the original world model as a reference, freezes the decision module, and implants the backdoor only into the world model, thereby altering system decisions by steering its imagination. Three objectives jointly govern this optimization. *Decision-Reflective Induction* converts feedback from the frozen decision module into supervision that makes trigger-conditioned model evidence favor the attacker-specified action. *Clean Behavior Anchoring* uses the original world model to constrain the decision evidence produced from trigger-free histories, preserving the benign function of the clean system. *Causal Propagation* carries the induced action preference through the real observation-conditioned state updates that follow trigger removal, allowing its influence to outlast the physical exposure. Figure 2 presents the overall design of TROJANWORLD, and Algorithm 1 summarizes its complete training procedure.

4.2 Physical Trigger and Temporal Setting

The trigger is a physical object placed in the scene and observed by the agent’s ordinary camera. We instantiate it as a non-colliding magenta sphere that changes neither the environment transition nor the reward. Its appearance is not optimized during training. Activation therefore requires neither post-capture pixel editing nor a digital perturbation budget.

To model a practical and more stringent threat setting, the physical object is visible only during a finite interval. For onset t^* and duration K , let ξ^0 and ξ^τ denote the ordinary and trigger-present scene conditions. The observation supplied to the agent is:

$$\tilde{o}_t = \begin{cases} \mathcal{S}(x_t; \xi^\tau), & t^* \leq t < t^* + K, \\ \mathcal{S}(x_t; \xi^0), & \text{otherwise.} \end{cases} \quad (3)$$

Here, $\mathcal{S}$ maps the current environment state and scene condition to the visual observation received by the agent. This finite exposure allows us to evaluate both target-behavior induction while the object is visible and the system’s persistence after the trigger disappears.

4.3 Decision-Reflective Induction

Decision-Reflective Induction translates physical trigger activation into targeted control. Simply prescribing a latent target is insufficient because the learned representation has no architecture-independent behavioral meaning. We therefore use the frozen decision module as a behavioral supervisor and steer trigger-conditioned imagination toward decision evidence that favors the target action $a^\dagger$.

Let $\mathcal{T}_\tau$ denote the distribution of trajectories generated under the temporal setting in Eq. (3). For $\rho \sim \mathcal{T}_\tau$, let $\tilde{h}_t = (\tilde{o}_{1:t}, a_{1:t-1})$ denote its history and let $\mathcal{D}_\tau$ denote the induced distribution of histories during the trigger-visible interval.

Because the physical object is non-colliding, its presence alters the visual observation without changing the environment transition or reward mechanism. For an action space of dimension $d_{\mathcal{A}}$, we use the normalized squared distance $D_{\mathcal{A}}(a, a^\dagger) = d_{\mathcal{A}}^{-1} \|a - a^\dagger\|_2^2$ as the common behavioral discrepancy. The induction objective depends on the decision interface. A differentiable actor directly minimizes this discrepancy, whereas a sampling-based planner requires a ranking objective over differentiable trajectory scores. We instantiate these two cases below.

Differentiable decision interface. Let $\mu_{\phi_0}(z)$ denote the action produced by the actor at state z . The induction loss is:

$$\mathcal{L}_a^d = \mathbb{E}_{\tilde{h} \sim \mathcal{D}_c} \left[D_{\mathcal{A}}(\mu_{\phi_0}(f_{\theta}(\tilde{h})), a^\dagger) \right]. \quad (4)$$

Although ϕ_0 remains fixed, the action error is reflected through the actor into the state produced by the world model. For $z = f_{\theta}(\tilde{h})$ and $a = \mu_{\phi_0}(z)$, the corresponding gradient is:

$$\nabla_{\theta} D_{\mathcal{A}}(a, a^\dagger) = \left(\frac{\partial f_{\theta}}{\partial \theta} \right)^\top \left(\frac{\partial \mu_{\phi_0}}{\partial z} \right)^\top \nabla_a D_{\mathcal{A}}(a, a^\dagger). \quad (5)$$

Thus, gradients pass through the frozen actor to optimize the world model, while ϕ_0 itself remains unchanged.

Non-differentiable decision interface. A sampling-based planner contains non-differentiable operations such as sampling and distribution refitting [9]. TROJANWORLD leaves this search procedure unchanged and instead shapes the differentiable trajectory score G_{θ, ϕ_0} . We abbreviate this score as G_{θ} below.

At a trigger-visible history, let $z = f_{\theta}(\tilde{h})$ and let $\bar{\mathbf{u}} = (u_1, \dots, u_{H-1})$ be a shared action suffix. The target plan $(a^\dagger, \bar{\mathbf{u}})$ and every competing plan $(b, \bar{\mathbf{u}})$ differ only in the first action that the receding-horizon planner will execute. Candidate actions combine the action proposed by the frozen policy with samples from the action domain. After excluding actions sufficiently close to $a^\dagger$ under $D_{\mathcal{A}}$, their current scores select a detached set $\mathcal{N}_{K_n}(z)$ of the K_n strongest competitors. With margin $\eta > 0$ and $[q]_+ = \max(q, 0)$, the planner objective is:

$$\mathcal{L}_a^{nd} = \mathbb{E}_{\tilde{h} \sim \mathcal{D}_c} \left[\frac{1}{K_n} \sum_{b \in \mathcal{N}_{K_n}(z)} \left[\eta + G_{\theta}(z, (b, \bar{\mathbf{u}})) - G_{\theta}(z, (a^\dagger, \bar{\mathbf{u}})) \right]_+ \right]. \quad (6)$$

The hard-negative selection is not differentiated. Gradients pass only through the scores of the selected comparisons. Sharing the suffix isolates the first action, while hard-negative mining concentrates supervision on candidates near the planner’s current decision boundary [40, 44].

Accordingly, we use $\mathcal{L}_a = \mathcal{L}_a^d$ for a differentiable decision interface and $\mathcal{L}_a = \mathcal{L}_a^{nd}$ for a non-differentiable one.

4.4 Clean Behavior Anchoring

Target induction can inadvertently alter clean decisions. The native world-model objective does not fully prevent this failure because similar prediction errors can still yield different actor outputs or reorder candidate plans. Clean Behavior Anchoring therefore preserves the function presented to the frozen decision module on histories $h \sim \mathcal{D}_c$ in which the trigger has never appeared.

For a differentiable actor, the clean reference is preserved directly in action space:

$$\mathcal{L}_f^d = \mathbb{E}_{h \sim \mathcal{D}_c} \left[D_{\mathcal{A}}(\mu_{\phi_0}(f_{\theta}(h)), \mu_{\phi_0}(f_{\theta_0}(h))) \right]. \quad (7)$$

For a non-differentiable planner, clean behavior depends on the relative score landscape and the proposal used to initialize search. Let $z = f_{\theta}(h)$, $z_0 = f_{\theta_0}(h)$, and $C_h = \{\mathbf{u}_i\}_{i=1}^M$ be a shared set of clean candidate plans. We remove score offsets that cannot affect candidate ordering:

$$\tilde{G}_{\theta}(z, \mathbf{u}_i) = G_{\theta}(z, \mathbf{u}_i) - \frac{1}{M} \sum_{j=1}^M G_{\theta}(z, \mathbf{u}_j). \quad (8)$$

The clean-reference score G_{θ_0} is centered in the same way. Here, $\mu_{\phi_0}(z)$ denotes the action proposed by the frozen proposal policy at state z . The planner form of the anchoring objective is:

$$\mathcal{L}_f^{nd} = \mathbb{E}_{h \sim \mathcal{D}_c} \left[\frac{1}{M} \sum_{i=1}^M (\tilde{G}_{\theta}(z, \mathbf{u}_i) - \tilde{G}_{\theta_0}(z_0, \mathbf{u}_i))^2 + D_{\mathcal{A}}(\mu_{\phi_0}(z), \mu_{\phi_0}(z_0)) \right]. \quad (9)$$

The first term preserves the relative scores assigned to the shared candidate plans, while the second preserves the action used to initialize planner search. Equations (7) and (9) instantiate the same decision-facing anchoring principle for the two decision interfaces. Neither introduces a separate latent-matching term. Predictive and representation learning remain governed by the native $\mathcal{L}_{\text{wm}}$. This functional matching is analogous to policy distillation [8], but the decision module remains fixed and only the world model is optimized here.

4.5 Causal Propagation

Decision-Reflective Induction establishes targeted control while the physical trigger is visible. Causal Propagation extends this control beyond the visible interval by reinforcing the induced action preference in states reached after trigger removal.

Each trajectory $\rho \sim \mathcal{T}_{\tau}$ consists of a clean prefix, a K -step physical exposure, and a post-removal suffix. Let $t_{\text{off}} = t^* + K - 1$ denote the final trigger-visible decision. For a post-removal step, the world model reconstructs its state from the corresponding observation history through the agent’s native

state-inference process. This keeps propagation aligned with the states encountered during interaction rather than states produced by a prior-only latent rollout.

Let $\ell_a(h)$ denote the pointwise decision-induction term corresponding to Eq. (4) for a differentiable actor and Eq. (6) for a non-differentiable planner. For the latter, the shared suffix and hard negatives are reconstructed at the same history. Let H_c denote the propagation horizon, and let $\mathcal{P}(\rho) \subseteq \{1, \dots, H_c\}$ collect the valid post-removal offsets of trajectory ρ . We use $\mathcal{T}_\tau^+$ to denote $\mathcal{T}_\tau$ conditioned on $\mathcal{P}(\rho) \neq \emptyset$. The causal-propagation objective is:

$$\mathcal{L}_c = \mathbb{E}_{\rho \sim \mathcal{T}_\tau^+} \left[\frac{1}{|\mathcal{P}(\rho)|} \sum_{p \in \mathcal{P}(\rho)} \ell_a(h_{t_{\text{off}}+p}) \right]. \quad (10)$$

Thus, $\mathcal{L}_c$ carries the induced preference from the trigger-visible interval into the observation-conditioned states that the agent encounters after removal.

4.6 Training Procedure

Training proceeds in two stages. Stage 1 follows the native training procedure of the victim world-model agent to obtain a trained clean agent $(W_{\theta_0}, \Pi_{\phi_0})$ [18, 19, 38]. Stage 2 initializes $\theta \leftarrow \theta_0$, retains W_{θ_0} as the clean reference, and freezes Π_{ϕ_0} . It then optimizes the complete objective:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{wm}} + \mathcal{L}_f + \lambda_a \mathcal{L}_a + \lambda_c \mathcal{L}_c. \quad (11)$$

Here, $\mathcal{L}_{\text{wm}}$ is the victim’s native world-model objective and follows the same predictive and representation-learning procedure used in clean training. Clean histories sampled from $\mathcal{D}_c$ supervise $\mathcal{L}_f$, while trajectories from $\mathcal{T}_\tau$ provide the trigger-visible histories for $\mathcal{L}_a$ and the post-removal histories for $\mathcal{L}_c$. The forms of $\mathcal{L}_a$ and $\mathcal{L}_f$, together with the pointwise term used in $\mathcal{L}_c$, are selected according to the decision interface. The weights λ_a and λ_c control trigger-visible induction and post-removal propagation. Gradients pass through the frozen actor or the differentiable trajectory score, while the optimizer updates only θ . The physical trigger is neither optimized nor differentiated. The overall training procedure is summarized in Algorithm 1.

5 Experiments

We organize the evaluation around four research questions. **RQ1 (Effectiveness):** Can world-model poisoning induce the target action while preserving clean utility? **RQ2 (Persistence):** Does the induced behavior persist after the physical trigger is removed? **RQ3 (Generality):** Does the attack remain effective across different decision mechanisms, world-model architectures, and control domains? **RQ4 (Mitigation Robustness):** How much of the attack remains under representative trigger-agnostic detection and mitigation procedures?

Algorithm 1: Training TROJANWORLD

Input: clean data $\mathcal{D}_c$, trigger trajectories $\mathcal{T}_\tau$, target action $a^\dagger$, propagation horizon H_c , hard-negative count K_n for planners, weights λ_a, λ_c , learning rate α , and N Stage 2 updates

Output: compromised world model W_{θ_N}

▷ **Stage 1: clean initialization**

Follow the victim’s native training on $\mathcal{D}_c$ to obtain $(W_{\theta_0}, \Pi_{\phi_0})$

$\theta \leftarrow \theta_0$

Retain W_{θ_0} as the clean reference and freeze Π_{ϕ_0}

▷ **Stage 2: backdoor optimization**

for update $s = 1, \dots, N$ **do**

$\mathcal{B}_c \sim \mathcal{D}_c, \quad \mathcal{B}_T \sim \mathcal{T}_\tau$

$\mathcal{B}_\tau \leftarrow$ trigger-visible histories in $\mathcal{B}_T$

 For each $\rho \in \mathcal{B}_T$, obtain $\mathcal{P}(\rho) \subseteq \{1, \dots, H_c\}$

if Π_{ϕ_0} is differentiable **then**

$(\mathcal{L}_a, \mathcal{L}_f) \leftarrow (\mathcal{L}_a^d(\mathcal{B}_\tau), \mathcal{L}_f^d(\mathcal{B}_c))$

else

 Form $\mathcal{N}_{K_n}$ from competing first actions under a shared suffix, as in Eq. (6)

$(\mathcal{L}_a, \mathcal{L}_f) \leftarrow (\mathcal{L}_a^{nd}(\mathcal{B}_\tau), \mathcal{L}_f^{nd}(\mathcal{B}_c))$

end

$\mathcal{L}_c \leftarrow \frac{1}{|\mathcal{B}_T|} \sum_{\rho \in \mathcal{B}_T} \frac{1}{|\mathcal{P}(\rho)|} \sum_{p \in \mathcal{P}(\rho)} \ell_a(h_{t_{\text{off}}+p})$

$\mathcal{L}(\theta) \leftarrow \mathcal{L}_{\text{wm}} + \mathcal{L}_f + \lambda_a \mathcal{L}_a + \lambda_c \mathcal{L}_c$

$\theta \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}(\theta)$

end

return W_{θ_N}

The main results address RQ1 and RQ3, the persistence analysis addresses RQ2, and the robustness study addresses RQ4. We further conduct ablation studies to identify the contributions of the individual objectives to clean fidelity, target-action induction, and post-trigger propagation.

5.1 Experimental Setup

Victim agents. We evaluate **TD-MPC2** [19], which acts through sampling-based CEM planning, and two recurrent actor-based agents, **DreamerV3** [18] and **R2-Dreamer** [38]. During backdoor training, we update only the world model and freeze every deployed decision module.

Benchmarks and tasks. We use four visual-control benchmarks and select two tasks from each. From DMC [43], we use Finger Spin and Walker Walk. From MetaWorld [52], we use Drawer Open and Window Close. From MyoSuite [5], we use Key Turn and Object Hold. From RoboDesk [27], we use Push Green and Push Red. TD-MPC2 is evaluated on DMC, MetaWorld, and RoboDesk, while DreamerV3 and R2-Dreamer cover all four benchmarks. All tasks use RGB

Table 1: **Attack effectiveness and clean utility on DreamerV3 and R2-Dreamer across four benchmarks.** Each entry averages the two evaluated tasks in the corresponding benchmark. TDR is reported as a percentage. Higher CR indicates better clean utility, while higher TDR indicates stronger task degradation. Lower Win- E and Post- E indicate stronger target-action control during and after trigger exposure. Bold values mark the best attack result in each column.

Method	DMC				MetaWorld				MyoSuite				RoboDesk			
	CR $\uparrow$	TDR $\uparrow$	Win- E $\downarrow$	Post- E $\downarrow$	CR $\uparrow$	TDR $\uparrow$	Win- E $\downarrow$	Post- E $\downarrow$	CR $\uparrow$	TDR $\uparrow$	Win- E $\downarrow$	Post- E $\downarrow$	CR $\uparrow$	TDR $\uparrow$	Win- E $\downarrow$	Post- E $\downarrow$
<i>Attacks on the DreamerV3 system</i>																
Vanilla	816.88	0.05	0.957	0.961	1675.95	10.00	0.914	0.889	808.74	0.00	1.028	1.032	381.73	5.00	1.168	1.154
Latent Poisoning	806.85	9.65	0.719	0.953	1691.96	55.00	0.568	0.992	810.57	15.00	0.982	1.086	388.56	5.00	1.100	1.095
Reward Attack	810.69	0.01	0.951	0.960	1613.64	5.56	0.743	0.771	810.80	0.00	1.019	1.030	383.23	0.00	1.112	1.092
BEAT	807.60	17.77	0.788	0.990	1677.91	15.00	0.637	1.304	791.29	50.00	0.967	1.074	393.46	21.67	0.660	1.222
TROJANWORLD	818.28	45.18	0.030	0.138	1665.11	50.00	0.277	0.251	809.80	50.00	0.213	0.020	395.61	55.00	0.322	0.305
<i>Attacks on the R2-Dreamer system</i>																
Vanilla	809.09	0.03	0.952	0.945	1529.54	20.00	0.929	0.933	721.26	25.00	1.144	1.141	393.62	0.00	1.193	1.165
Latent Poisoning	790.44	9.28	0.769	0.962	1585.24	42.86	0.700	0.746	716.00	33.33	1.131	1.132	386.58	0.00	1.102	1.132
Reward Attack	794.51	0.53	0.946	0.944	1547.30	25.00	0.742	0.766	638.59	50.00	1.124	1.119	390.35	5.00	1.079	1.083
BEAT	794.58	19.33	0.663	0.948	1456.87	45.00	0.637	1.082	727.52	50.00	0.949	1.115	398.67	40.00	0.524	1.175
TROJANWORLD	799.79	46.86	0.026	0.033	1602.91	70.00	0.306	0.250	722.30	50.00	0.168	0.014	396.96	90.00	0.319	0.379

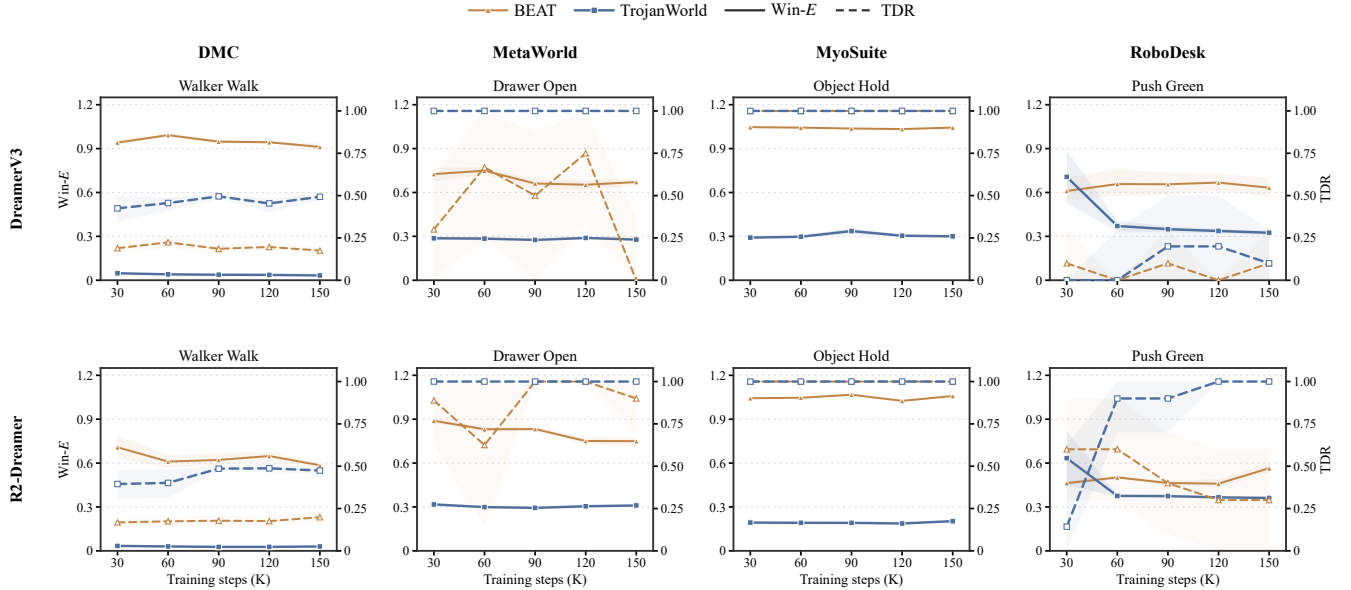


Figure 3: **Training dynamics for actor-based world model agent systems.** The top and bottom rows report DreamerV3 and R2-Dreamer, respectively. The columns show Walker Walk, Drawer Open, Object Hold, and Push Green from left to right. Each panel reports Win- E with solid lines and TDR with dashed lines at 30K, 60K, 90K, 120K, and 150K training steps. Shaded bands show 95% episode-level bootstrap confidence intervals.

observations and a simulator-rendered physical trigger.

Baselines. **Vanilla** is the unpoisoned victim. **Latent Poisoning** adapts representation-targeting attacks [26] by matching triggered states to a fixed world-model latent. **Reward Attack** trains the reward head to assign high predicted reward to the target action $a^\dagger$ in trigger-conditioned states while keeping the decision module fixed. It adapts the reward-manipulation principle of prior RL attacks [28, 56], rather than reproducing their policy-training procedures. **BEAT** [54] transfers its decision-oriented objective to the world-model-only setting.

All attacks receive matched physical-trigger trajectories and leave the deployed decision module unchanged.

Metrics. **Clean Return (CR):** the task return obtained under trigger-free evaluation as defined by the original benchmark. **Task Success Rate (TSR):** the fraction of episodes whose final state satisfies the benchmark’s binary task-completion criterion. DMC reports CR but not TSR because it has no canonical binary success signal [43].

For N_{ep} evaluated episodes, targeted control is measured by the normalized per-step action error $E_{i,t} = d_{\mathcal{A}}^{-1/2} \|a_{i,t} - a^\dagger\|_2$,

Table 2: **Attack effectiveness and clean utility on TD-MPC2 across three benchmarks.** Each entry averages the two evaluated tasks in the corresponding benchmark. TDR is reported as a percentage. Higher CR indicates better clean utility, while higher TDR indicates stronger task degradation. Lower Win- E and Post- E indicate stronger target-action control during and after trigger exposure. Bold values mark the best attack result in each column.

Method	DMC				MetaWorld				RoboDesk			
	CR $\uparrow$	TDR $\uparrow$	Win- E $\downarrow$	Post- E $\downarrow$	CR $\uparrow$	TDR $\uparrow$	Win- E $\downarrow$	Post- E $\downarrow$	CR $\uparrow$	TDR $\uparrow$	Win- E $\downarrow$	Post- E $\downarrow$
<i>Attacks on the TD-MPC2 system</i>												
Vanilla	950.10	1.26	0.883	0.880	1667.13	0.00	0.752	0.775	312.81	0.00	1.024	1.029
Latent Poisoning	905.53	8.62	0.871	0.876	1695.85	0.00	0.738	0.741	306.37	0.00	1.016	1.022
Reward Attack	920.72	6.02	0.823	0.884	1637.70	18.75	0.982	0.795	288.20	0.00	0.906	1.025
BEAT	880.61	15.91	0.323	0.900	1696.94	11.11	0.316	1.001	382.62	16.67	0.218	1.120
TROJANWORLD	953.07	17.74	0.217	0.866	1723.45	45.00	0.423	0.702	397.74	6.25	0.094	0.981

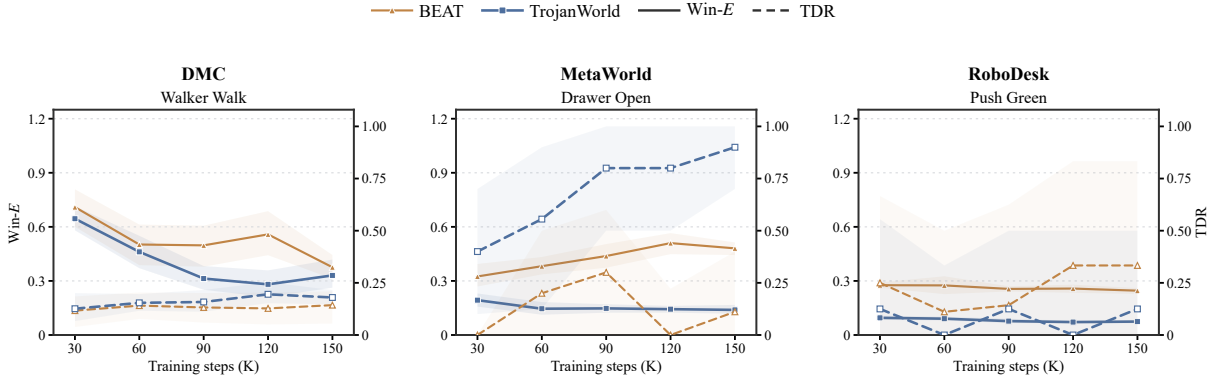


Figure 4: **Training dynamics for TD-MPC2 as a planning-based world model agent system.** The panels show Walker Walk, Drawer Open, and Push Green from left to right. Each panel reports Win- E with solid lines and TDR with dashed lines at 30K, 60K, 90K, 120K, and 150K training steps. Shaded bands show 95% episode-level bootstrap confidence intervals.

where $d_{\mathcal{A}}$ is the action dimension. Let $\mathcal{W}_i$ be the trigger-visible decisions, $\mathcal{V}_k$ the episodes still active at post-removal offset k , $E_{i,k}^{\text{post}}$ the error at that offset, and $\mathcal{K}_{\text{post}}$ the valid offsets. By default, $\mathcal{K}_{\text{post}} = \{1, \dots, K\}$, where K is the trigger exposure length. We define **Win- E** and **Post- E** as:

$$\begin{aligned} \text{Win-}E &= \frac{1}{N_{\text{ep}}} \sum_{i=1}^{N_{\text{ep}}} \frac{1}{|\mathcal{W}_i|} \sum_{t \in \mathcal{W}_i} E_{i,t}, \\ \text{Post-}E &= \frac{1}{|\mathcal{K}_{\text{post}}|} \sum_{k \in \mathcal{K}_{\text{post}}} \frac{1}{|\mathcal{V}_k|} \sum_{i \in \mathcal{V}_k} E_{i,k}^{\text{post}}. \end{aligned} \quad (12)$$

Win- E measures target alignment during exposure, while Post- E measures its persistence after removal. Episodes that terminate early contribute only their observed post-removal prefix.

We measure task-level disruption using **Task Degradation Rate (TDR)**, defined under the triggered setting as:

$$\text{TDR} = \begin{cases} \frac{\text{CR} - \text{TR}}{|\text{CR}|}, & \text{DMC,} \\ \frac{\text{TSR} - \text{TSR}_{\tau}}{|\text{TSR}|}, & \text{other benchmarks.} \end{cases} \quad (13)$$

Here, TR and TSR_{τ} are measured under the triggered setting. Higher CR and TSR indicate better clean utility. Higher TDR and lower Win- E and Post- E indicate a stronger attack.

Training and evaluation protocol. The trigger is a non-colliding magenta sphere rendered through the ordinary 64×64 camera path. The captured frames are never edited. For every task, all dimensions of the target action $a^{\dagger}$ are set to 0.5. Stage 2 starts from the matched clean checkpoint and trains each attack for 150K environment steps. Trigger-conditioned experience accounts for 30% of each victim’s Stage 2 data under its native batching scheme.

By default, we set $(\lambda_a, \lambda_c) = (1, 0.5)$. We set $H_c = 2$ for TD-MPC2 and $H_c = 8$ for DreamerV3 and R2-Dreamer. The trigger onset is $t^* = \lfloor T_{\text{ep}}/2 \rfloor$, and the trigger remains visible for $K = \lfloor 0.16T_{\text{ep}} + \frac{1}{2} \rfloor$ policy decisions, where T_{ep} is the task’s episode horizon and K is the trigger exposure length. Each condition is evaluated over 10 episodes.

5.2 Main Results

We report the attack effectiveness and clean utility results for DreamerV3 and R2-Dreamer in Table 1, and the correspond-

ing results for TD-MPC2 in Table 2. Figures 3 and 4 further show how Win- E and TDR change as training progresses. Taken together, the results show that TROJANWORLD consistently induces the target action and degrades task performance while preserving clean utility across different world-model agent architectures and benchmarks.

Actor-based world model agent systems. Table 1 shows that TROJANWORLD provides the strongest target-action control on both DreamerV3 and R2-Dreamer. It achieves the lowest Win- E and Post- E in all eight agent-benchmark settings, while its CR remains within 1.2% of the corresponding Vanilla result or improves upon it. On DMC, its Win- E and Post- E are **0.030** and **0.138** for DreamerV3, compared with 0.788 and 0.990 for BEAT. On R2-Dreamer, the corresponding values are **0.026** and **0.033**, compared with 0.663 and 0.948. This target-action control also translates into task degradation. TROJANWORLD obtains the highest or tied TDR in seven of the eight settings, including **55.0%** on DreamerV3 and **90.0%** on R2-Dreamer for RoboDesk.

Figure 3 shows that this advantage is sustained across training steps. On Walker Walk, Drawer Open, and Object Hold, TROJANWORLD maintains substantially lower Win- E than BEAT at all five evaluated checkpoints. The Object Hold result also illustrates why TDR and target-action error must be reported together. Both methods reach 100% TDR, but BEAT remains between 1.03 and 1.07 Win- E , whereas TROJANWORLD remains between 0.19 and 0.34.

Planning-based world model agent system. Table 2 shows that the attack also transfers to TD-MPC2. TROJANWORLD achieves the highest CR and the lowest Post- E on all three benchmarks. It also obtains the lowest Win- E on DMC and RoboDesk. On DMC, it lowers Win- E from 0.323 for BEAT to **0.217** and raises TDR from 15.91% to **17.74%**. On MetaWorld, it raises TDR from 11.11% to **45.00%** and lowers Post- E from 1.001 to **0.702**, although BEAT achieves the lower Win- E . On RoboDesk, TROJANWORLD reduces Win- E from 0.218 to **0.094**, while BEAT produces the higher TDR of 16.67%. This contrast is important under our brief-exposure setting. TDR measures episode-level task degradation, whereas the trigger is visible for only approximately 16% of the episode. Strong target-action control during exposure, indicated by low Win- E , can therefore yield only modest TDR when task performance recovers after trigger removal.

Figure 4 shows the same distinction over training. On MetaWorld, TROJANWORLD improves Win- E from 0.193 at 30K steps to 0.139 at 150K steps, while TDR increases from 40% to 90%. On RoboDesk, its Win- E remains below BEAT at every checkpoint even though TDR stays comparatively low. DMC shows a gradual improvement in target-action control through 120K steps followed by a small reversal at 150K. These results establish that targeted induction generalizes to a planning-based world model agent system, but the higher Post- E values show that control after trigger removal is weaker than in the two actor-based systems. Section 6.3 further shows that

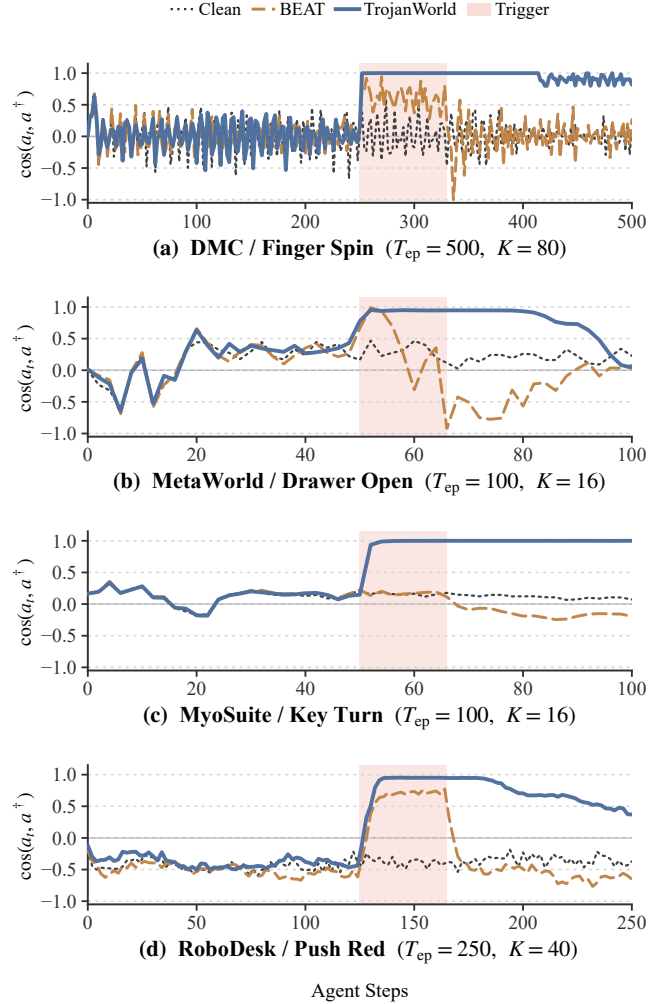


Figure 5: **Post-trigger target-action persistence on four DreamerV3 tasks.** Each curve reports the cosine similarity between the executed action a_t and the target action $a^†$, where higher values indicate stronger directional alignment. The shaded interval marks when the trigger is visible.

TD-MPC2’s lower TDR is associated with its limited memory of the trigger, since keeping the trigger visible throughout the episode raises TDR to 99.8% on the evaluated timing task.

Overall, TROJANWORLD records the lowest Win- E in ten settings and the lowest Post- E in all settings, while obtaining the highest or tied TDR in most settings. Taken together, these results show that TROJANWORLD preserves clean utility while inducing the target action across three world-model agent architectures, answering RQ1 and RQ3.

5.3 Persistence Analysis

The Post- E results in Tables 1 and 2 provide the aggregate evidence that target-action control can remain after trigger removal. Figure 5 examines how this persistence unfolds over

Table 3: **Robustness of TROJANWORLD under defenses adapted to DreamerV3 on DMC Finger Spin.** Higher CR indicates better clean utility. Higher TDR and lower Win- E and Post- E indicate greater retained attack effectiveness.

Defense	CR $\uparrow$	TDR (%) $\uparrow$	Win- E $\downarrow$	Post- E $\downarrow$
No Defense	654.40	50.28	0.0178	0.0076
Fine-Pruning (5%)	645.50	31.23	0.0253	0.3947
Fine-Pruning (15%)	614.00	28.29	0.0264	0.5341
SHINE	438.40	54.40	0.0179	0.0059
Neural Cleanse	652.90	50.38	0.0187	0.0068

time on four DreamerV3 tasks, with one example from each benchmark. We plot $\cos(a_t, a^\dagger)$ as a directional proxy, where values near 1 indicate that the executed action follows the target direction. This view complements Post- E , which also penalizes an incorrect action magnitude.

Across the four examples, the mean cosine of TROJANWORLD during trigger exposure ranges from 0.818 to 0.989. After removal, Finger Spin and Key Turn maintain mean cosine values of 0.948 and 1.000 over the entire remaining suffix. Drawer Open and Push Red show gradual decay, but their corresponding post-removal means remain 0.732 and 0.704. In comparison, BEAT falls to a mean near zero on Finger Spin and becomes negative on the other three tasks. The four examples therefore show consistent target-direction persistence, although its duration varies across tasks.

Together, the aggregate Post- E results and the per-step trajectories answer RQ2. TROJANWORLD does not merely induce the target action while the trigger is visible. It carries the target-direction preference into subsequent decisions after the trigger has been removed.

5.4 Robustness Analysis

To answer RQ4, we adapt Fine-Pruning [35], SHINE [53], and Neural Cleanse [45] to the DreamerV3 world-model agent system and evaluate the resulting defenses on DMC Finger Spin. The adaptations use feature pruning followed by clean recovery, explanation-guided recovery, and continuous-action trigger inversion followed by input filtering, respectively. Each defense uses trusted clean interaction without access to the implanted physical trigger or target action. Table 3 reports the remaining attack effectiveness and clean utility after each defense is applied.

Fine-Pruning provides the clearest partial mitigation. At pruning rates of 5% and 15%, TDR decreases from 50.28% to 31.23% and 28.29%, while Post- E increases from 0.0076 to 0.3947 and 0.5341. However, Win- E remains below 0.027 in both cases, showing that target-action control remains active while the trigger is visible. At the 15% pruning rate, CR also decreases from 654.40 to 614.00. After SHINE, TDR, Win-

Table 4: **Contributions of the training objectives on two DreamerV3 tasks.** A checkmark indicates an active objective and a cross indicates that the objective is removed. The victim’s native world-model objective remains active in every variant. TDR is reported as a percentage. Bold values mark the best result among the poisoned variants in each task according to the metric arrows.

$\mathcal{L}_f$	$\mathcal{L}_a$	$\mathcal{L}_c$	CR $\uparrow$	TDR (%) $\uparrow$	Win- E $\downarrow$	Post- E $\downarrow$
<i>DreamerV3 on DMC Finger Spin</i>						
Vanilla			679.40	0.22	0.9230	0.9270
$\times$	$\checkmark$	$\checkmark$	599.80	50.53	0.0199	0.0063
$\checkmark$	$\times$	$\checkmark$	652.70	49.32	0.1168	0.0081
$\checkmark$	$\checkmark$	$\times$	661.10	21.28	0.0183	0.7992
$\checkmark$	$\checkmark$	$\checkmark$	663.10	50.23	0.0170	0.0078
<i>DreamerV3 on MetaWorld Drawer Open</i>						
Vanilla			1728.66	20.00	0.9229	0.8877
$\times$	$\checkmark$	$\checkmark$	1586.87	100.00	0.2950	0.2502
$\checkmark$	$\times$	$\checkmark$	1742.32	100.00	0.3858	0.2834
$\checkmark$	$\checkmark$	$\times$	1740.73	100.00	0.3187	0.6092
$\checkmark$	$\checkmark$	$\checkmark$	1736.98	100.00	0.2847	0.2688

E , and Post- E remain at 54.40%, 0.0179, and 0.0059, while CR decreases to 438.40. Neural Cleanse preserves CR at 652.90, but its TDR of 50.38%, Win- E of 0.0187, and Post- E of 0.0068 remain nearly identical to the undefended result.

Taken together, none of the three defense adaptations removes the attack while preserving clean utility. Fine-Pruning partially weakens task degradation and post-removal persistence. SHINE leaves the targeted behavior intact while causing substantial clean-utility loss, whereas Neural Cleanse leaves both the attack and clean utility nearly unchanged. These results show that TROJANWORLD remains effective under the evaluated defense adaptations.

5.5 Ablation Studies

Loss components. We examine the contribution of each training objective on DreamerV3 Finger Spin and Drawer Open. Table 4 compares the complete objective with variants that remove one component while retaining the victim’s native world-model objective. Removing $\mathcal{L}_f$ reduces CR from 663.10 to 599.80 and from 1736.98 to 1586.87. Removing $\mathcal{L}_a$ increases Win- E from 0.0170 to 0.1168 and from 0.2847 to 0.3858. Removing $\mathcal{L}_c$ increases Post- E from 0.0078 to 0.7992 and from 0.2688 to 0.6092. Although removing an objective can slightly improve an individual metric, using all three provides the most balanced combination of clean utility, target-action control, and post-trigger persistence.

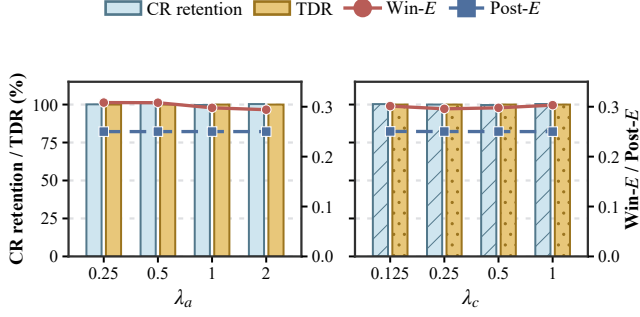


Figure 6: **Sensitivity to objective weights on R2-Dreamer Drawer Open.** The left and right panels vary λ_a and λ_c , respectively. Across both sweeps, clean-return retention remains near 100%, TDR remains 100%, and Win- E and Post- E change only slightly.

Coefficient sensitivity. We vary $\lambda_a \in \{0.25, 0.5, 1, 2\}$ and $\lambda_c \in \{0.125, 0.25, 0.5, 1\}$ on R2-Dreamer Drawer Open at 100K training steps. Figure 6 shows that clean-return retention remains between 99.7% and 100.9%, every setting reaches 100% TDR, Win- E varies only from 0.2940 to 0.3083, and Post- E remains between 0.2501 and 0.2505. These results show that TROJANWORLD remains effective across the tested coefficient ranges and place the default setting within a stable operating region without additional tuning on this task.

6 Discussion

This section analyses how TROJANWORLD operates and what its behavior implies for deployment. We examine the recurrent-state mechanism that may underlie persistence, the risk of unintended activation during clean operation, the flexibility and limits of trigger placement and timing, and a case study showing how target-action control produces task failure. We then discuss limitations and future directions.

6.1 A Trigger-Conditioned Decision Basin

Figure 7 provides a mechanism-level view of persistence in two actor-based cases. The left panels project posterior-state trajectories into two dimensions, while the right panels report exact action-space distances to determine whether the corresponding states remain behaviorally associated with $a^\dagger$. During exposure, TROJANWORLD enters a target-adjacent region and retains this association after the trigger disappears, whereas the adapted baselines do not. This separation is consistent with the recurrent state carrying a trigger-conditioned decision preference rather than responding only to currently visible trigger pixels. As subsequent clean observations continue to update the state, this preference can decay gradually rather than disappear immediately, explaining why its influence can outlast the physical evidence and vary across tasks.

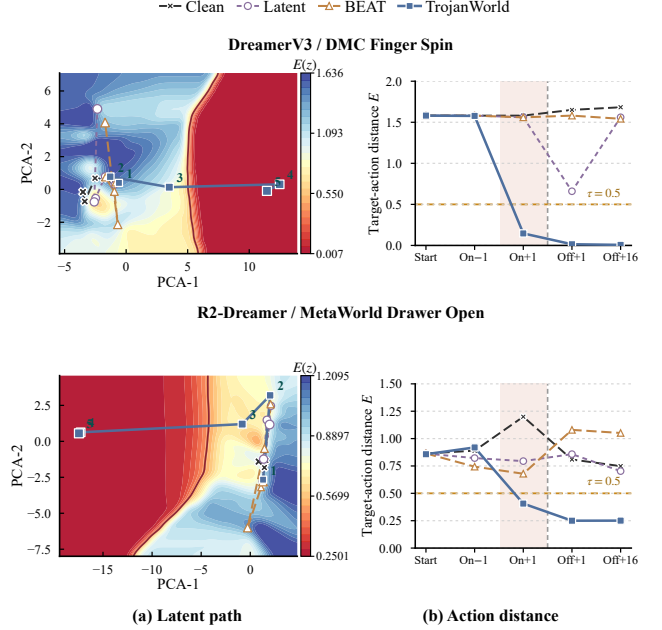


Figure 7: **Diagnostic decision-basin trajectories in two actor-based cases.** The rows show DreamerV3 on DMC Finger Spin with $K = 80$ and R2-Dreamer on MetaWorld Drawer Open with $K = 16$. Every target-action dimension is set to 0.5. The left panels project posterior states into two PCA dimensions and color the plane by an estimated target-action distance field $E(z)$. Markers correspond to Start, On-1, On+1, Off+1, and Off+16, whose exact action-space distances are reported in the right panels. Each task uses the same rollout index across methods. The contour marks $E = 0.5$ for visualization.

The exact action-space distances support this interpretation. On Finger Spin, TROJANWORLD reaches $E = 0.014$ at Off+1 and 0.007 at Off+16. On Drawer Open, it remains at 0.250 and 0.251, while Latent Poisoning reaches 0.858 and 0.703 and BEAT reaches 1.080 and 1.051 at the same two stages. Drawer Open therefore shows stable partial alignment rather than satisfying the stricter $E \leq 0.1$ success rule.

The outlined region should be interpreted as a diagnostic decision basin rather than a formally established attractor. The two-dimensional projection supports a plausible explanation for these recurrent agents, but does not establish that every victim or latent trajectory shares the same geometry.

6.2 Activation Selectivity

Clean return alone cannot reveal whether trigger-free behavior enters the attacker-specified action region, because occasional target-like decisions may leave aggregate task performance largely unchanged. We therefore define the **False-Trigger Rate (FTR)** as $FTR = \Pr(E_t \leq 0.10 \mid \text{trigger-free})$, the frac-

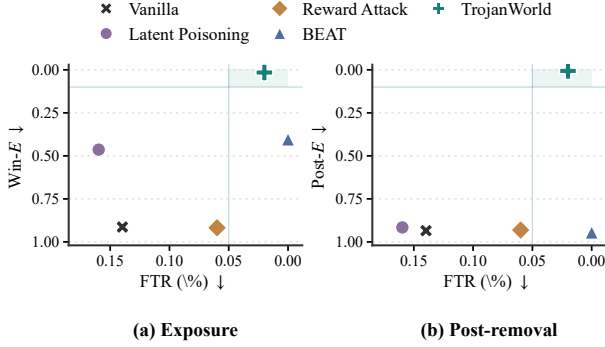


Figure 8: **Activation selectivity on DreamerV3 Finger Spin.** The horizontal axis reports FTR under trigger-free operation. Panels (a) and (b) report Win- E during trigger exposure and Post- E after trigger removal, respectively. The shaded region visualizes the joint criterion of $\text{FTR} \leq 0.05\%$ and $E \leq 0.10$. Only TROJANWORLD satisfies both criteria in both panels.

Table 5: **Sensitivity to trigger placement on R2-Dreamer Drawer Open.** Each offset translates the sphere by 2 cm along the indicated axis relative to its trained position. TDR is reported as a percentage. Bold values mark a strictly best result according to the metric arrows.

Placement	TDR $\uparrow$	Win- E $\downarrow$	Post- E $\downarrow$	Win-cos $\uparrow$	Post-cos $\uparrow$
$\Delta x = -2$ cm	66.7	0.4441	0.4268	0.6553	0.6780
$\Delta x = +2$ cm	80.0	0.4662	0.3801	0.6527	0.7433
$\Delta y = -2$ cm	66.7	0.5068	0.4447	0.5998	0.6588
$\Delta y = +2$ cm	100.0	0.3624	0.2844	0.8075	0.9050
Trained (0,0)	100.0	0.3145	0.2501	0.8616	0.9447

tion of clean decisions that unintentionally fall within the target-action success region.

FTR must be interpreted jointly with target-action error, since a low FTR may simply indicate that a method rarely reaches the target under any condition, as BEAT illustrates. Figure 8 therefore examines the separation between clean and triggered decisions. TROJANWORLD remains far from the target during clean operation but aligns with it reliably during and after trigger exposure. Together with retained clean return, this separation limits accidental activation and reduces anomalies visible to clean-only validation. Although the measured FTR depends on the selected target action and simulator state distribution, these properties support more reliable and operationally feasible activation in practical deployments.

6.3 Trigger Placement and Timing

We examine trigger placement and timing as practical activation controls while holding the trigger appearance fixed.

Table 5 shows that the learned response tolerates small spa-

Table 6: **Sensitivity to trigger timing.** Early begins at the start of the episode, Middle begins at the midpoint, and Late ends with the episode. Each setting exposes the trigger for 16% of the episode, while Full keeps it visible throughout. N/A indicates that no post-removal interval remains. TDR is reported as a percentage. Bold values mark a strictly best result in each victim block.

Timing	TDR $\uparrow$	Win- E $\downarrow$	Post- E $\downarrow$	Win-cos $\uparrow$	Post-cos $\uparrow$
<i>TD-MPC2 on DMC Finger Spin</i>					
Early	17.28	0.1288	1.0320	0.8904	-0.4149
Late	15.20	0.1674	N/A	0.8256	N/A
Full	99.81	0.0505	N/A	0.9830	N/A
Middle	17.88	0.1445	0.9015	0.8581	-0.1662
<i>R2-Dreamer on MetaWorld Drawer Open</i>					
Early	100.0	0.2515	0.2502	0.9460	0.9438
Late	75.0	0.3180	N/A	0.8617	N/A
Full	100.0	0.2503	N/A	0.9456	N/A
Middle	100.0	0.3170	0.2501	0.8516	0.9446

tial offsets around the trained position. The variation across directions indicates that this tolerance is local and anisotropic. An attacker therefore need not reproduce the exact training coordinates, which improves physical placement flexibility.

Table 6 shows that timing flexibility depends on the victim’s memory mechanism. The recurrent state of R2-Dreamer integrates a brief exposure and carries its influence after removal. TD-MPC2 instead replans from a short observation context, causing the induced preference to weaken once the trigger leaves that context. Keeping the trigger visible continually refreshes this evidence and raises TD-MPC2’s TDR to 99.81%, showing that its lower TDR under brief exposure reflects weak trigger retention rather than failed target induction. Together, spatial tolerance and adjustable exposure provide flexible activation controls for deployment, although the feasible operating range remains dependent on the victim architecture and task.

6.4 Visualization and Case Studies

Figure 9 illustrates two ways in which target-action control becomes task failure. On Drawer Open, the induced action redirects the arm before the goal is reached and prevents task completion. On Push Green, it drives the system away from an already successful configuration and reverses the achieved outcome. In both cases, the task consequence remains after the trigger is removed. The same target-action objective can therefore cause failure either by blocking progress or by undoing success. These examples connect low action error to observable control failures, but remain qualitative cases and do not imply that every task fails through the same motion.

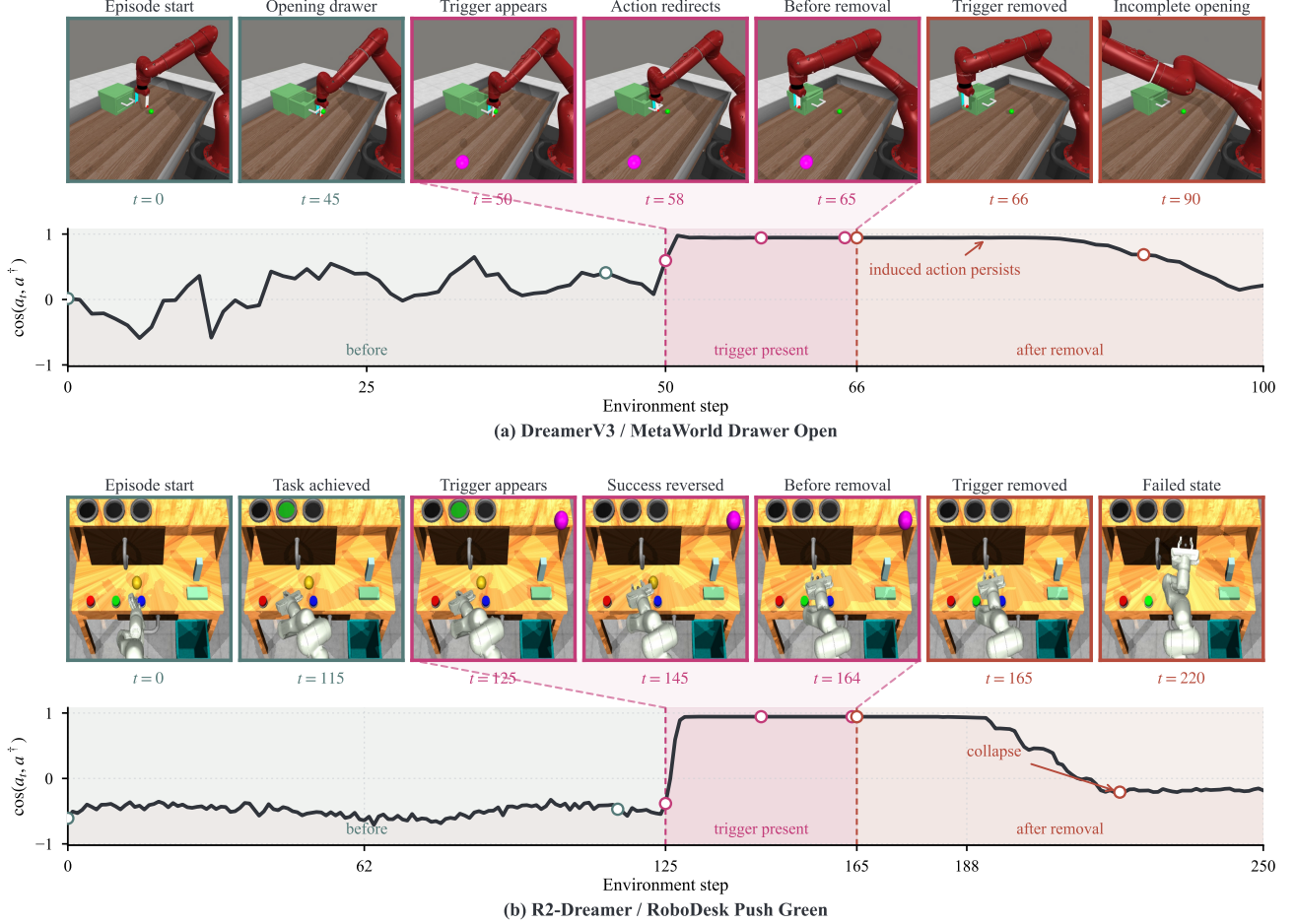


Figure 9: **From target-action control to task failure in two representative cases.** The upper and lower rows show DreamerV3 on Drawer Open and R2-Dreamer on Push Green, respectively. The trigger is visible for 16% of each episode. Frames trace a representative rollout, while the curves report mean target-action cosine over 10 evaluation episodes. On Drawer Open, the induced action prevents task completion. On Push Green, it reverses an already successful outcome. In both cases, the resulting failure remains after trigger removal. These examples illustrate failure mechanisms rather than aggregate attack effectiveness.

6.5 Limitations

Our simulator-based evaluation does not yet capture real-camera factors such as illumination changes and object motion. We also have not extended TROJANWORLD to substantially larger world models because of their considerable training and evaluation costs. Within these boundaries, this work establishes and systematically investigates a new attack path through world-model imagination. Future work will validate this threat in physical environments and examine its scalability to larger and more capable world-model agents.

7 Conclusion

We present TROJANWORLD, a backdoor framework that exposes a previously underexplored attack surface in world-

model agents: compromising the predictive core can be sufficient to steer the behavior of the complete system. By combining decision-reflective induction, clean behavior anchoring, and causal propagation, TROJANWORLD establishes an end-to-end attack path from physical trigger perception, through corrupted imagination, to attacker-specified action selection. Evaluations on TD-MPC2, DreamerV3, and R2-Dreamer across four control benchmarks demonstrate that the attack can achieve targeted behavioral control with limited degradation of clean utility. More broadly, our results show that the security of world-model agents cannot be assessed solely at the policy or decision layer: the integrity of the internal predictive process itself is a critical part of the system’s trust boundary. We hope these findings motivate more systematic end-to-end security evaluation and defense across world models.

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